

Research on Positive Emotion Recognition Based on EEG Signals

1st Xuan Chen

Faculty of Data Science
City University of Macau
Macau 999078, China
D21091100028@cityu.mo

2nd* WenJian Liu

Faculty of Data Science
City University of Macau
Macau 999078, China
andylau@cityu.mo

Abstract—Human logical decision-making, perception, learning, and many functions, which influence people's decision-making on things and perception of external things, are all significantly influenced by emotions. An essential component of emotional computing is emotional cognition. By examining the psychological and psychological traits of the client, it evaluates the psychological condition of the service object. Currently, most of the research on emotions based on electroencephalogram (EEG) signals focuses on classifying positive, neutral, and negative emotions, or studying negative emotions, with less attention paid to specifically identifying positive emotions. This experiment proposed an experimental design that utilized virtual reality technology as an inducing method to stimulate positive emotions, and identified and evaluated the emotions of happiness, desire, and healing. This experiment collected datasets on three positive emotions of happiness, desire, and healing, and collected the Positive Affect and Negative Affect Scale (PANAS) and self-assessment (SAM) form of the participants. Divided the experiment into two groups, one was the experimental group wearing VR, and the other was the experimental group not wearing VR. The immersion feeling scale was used to study the effect of VR on stimulating emotions. Through the design of emotion induction experiments, the EEG signals of experiencing happiness, desire, and healing were collected. The EEG signals were input into a CNN network for feature extraction, both in the form of images and time series. The Resnet18 network was used for image-based emotion recognition with an accuracy of 93%. The time-series data was processed using an LSTM network for emotion recognition with an accuracy of 94.9%.

Keywords—EEG; CNN; emotion recognition; feature extraction; LSTM

I. INTRODUCTION

In people's daily life, the transmission of emotion is indispensable. It can not only maintain people's information communication, but also become the fundamental guarantee for people to adapt and maintain their survival. According to the point of view of cognition, the human brain and emotions are closely related. Joy can make people happiness, anger can make people lose their minds, and sadness can make people sad; fear can make people feel afraid, and the appearance of these emotions will have a certain impact on our thinking, cognition and judgment. Emotional cognition is an important component of emotional processing. It determines the psychological state of the service object through the analysis of their psychological characteristics.

Long-term unpleasant emotional states can exacerbate psychiatric conditions and potentially put oneself and society in danger. However, a positive and excited emotional state can help improve learning and work efficiency, expand thinking and improve creativity, and can also improve body immunity. Therefore, timely and effective emotion recognition has a wide range of uses and great significance. In terms of nursing, you can monitor the mood of the elderly at any time, and cooperate with physiological parameters such as blood pressure, pulse, and respiration. Once you find something wrong, you can report to the nursing department to ensure their physical condition. Timely monitoring can prevent students' bad behaviors. At the same time, teachers can also discover students' psychological states and offer timely psychological treatment to those who are in poor mental health so that they can swiftly overcome their negative feelings and learn more effectively and quickly. In terms of medical diagnosis, for patients with symptoms such as depression, deaf-mute, and mania, different psychological changes can be made according to the patient's psychological state, so as to provide an objective and efficient evaluation method, so that the patient can recover as soon as possible.

At present, there are three methods for emotion recognition: the first method is emotion recognition based on non-physiological signals, which focuses on the analysis of facial expressions, voice intonation, and actions. This method does not require special equipment, but it is simple to conceal. After receiving relevant psychological training, it can well cover up or exaggerate one's true expression, tone and behavior. The second type of method focuses on the analysis of changes in peripheral autonomic nerves under various emotional conditions, such as electrocardiogram (ECG), Electrocardiogram (EMG) and heart rate pulse. Compared with the first type of method, the emotion recognition data provided by peripheral physiological signals can reflect relevant reference information more accurately and finely. However this method frequently lacks a suitable measurement. The third type of method is to use EEG to describe the movement of neurons from the center to the cerebral cortex, and record the changes in their potential. EEG can be used as a channel of communication between the human brain and emotions because it is intimately tied to emotions. Therefore, it has recently become a focus of research to examine emotion identification using EEG data. All in all, the second and third types of methods are collectively referred to as emotion

recognition methods based on physiological signals. Since physiological signals are difficult to disguise, these two types of methods are both objective and spontaneous.

The novel aspect of our research proposal was the distinction between the three positive emotions—happiness, desire, and healing. The sense of presence and immersion of the VR scene was used to stimulate the subjects to produce specific emotional responses, and the data sets of EEG signals were collected. The EEG signal was input in the form of a picture, and the Resnet18 network in CNN was directly used for feature extraction and emotion recognition.

II. RESEARCH STATUS OF EEG SIGNALS

A. Current status of research on methods of inducing emotional EEG signals

Successfully inducing various emotional states is a prerequisite for sentiment analysis. Nowadays, there exist databases for emotional physiology [1][2][3], particularly EEG-based emotional databases [1][3], and databases for emotional expression (such as facial expressions, postures, and voices) [1][4][5][6]. The number is quite little in comparison.

Siedlecka *et al.* [7] used five experimental induction techniques: visual stimuli, music, autobiography, situational programs, and imagery to induce the emotions of anger, disgust, surprise, happiness, fear, and sadness in the subjects.

Li *et al.* [8] stimulated the positive and negative emotions of the subjects by means of self-induction (recalling music clips after listening to music) and stimulus induction (listening to music), and monitored and collected the EEG signals of the subjects. Thus the data set of MSI was established.

Jaidee *et al.* [9] collected subjects' facial data sets while using audio and image data sets as stimulus materials to elicit the subjects' feelings of enjoyment, relaxation, neutrality, anger, and sadness.

Sanders koelstra *et al.* [10] established a multi-modal data set. In this experiment, listening to music evoked the emotions of the test subjects, and recorded the test subjects' EEG signals, physiological data and facial videos. The self-assessment form of the subjects' analyzed the degree of valance, arousal and liking. The DEAP dataset, which has been acknowledged by numerous studies and used by numerous research institutes, is made up of these two types of data.

A. Gruenewald *et al.* [11] constructed their own innovative emotion-inducing scenarios (happiness: reading a text with happiness while listening to classical music, boredom: watching a lecture video on sayings, frustration: being controlled by someone while playing a computer game). Stimulated the happiness, boredom, and frustration of the subjects, and at the same time collected the temperature, EEG, ECG, skin galvanic response, heart rate and blood oxygen saturation of the subjects.

Bos *et al.* [12] compared three different eliciting methods through three different stimulating materials: visual, auditory, and audio-visual. The results showed that the result induced by a single visual stimulating material was not ideal.

According to the analysis above, although the researchers used a single stimulation method to obtain the desired effect, the effect of a single simple emotion induction method (such as viewing images) was not obvious. Future research can combine multiple induction methods to stimulate the emotions of the subjects. There aren't enough emotional EEG data in the existing databases, neither in terms of quantity nor variety of participants, to support further study. The work of the DEAP database has currently been acknowledged by many scholars due to the quantity and variety of databases available, and numerous studies have also been conducted using the DEAP database.

In order to solve these problems, this experiment designed an experimental plan that used virtual reality technology as an inducing method to stimulate people's positive emotions by using the sense of presence and immersion in VR scenes, and collected the EEG signals, PANAS scale and self-assessment form of the subjects. Constituted the database belonging to this experiment. Finally, identified and evaluated the emotions of happiness, desire, and healing.

B. Research Status of EEG Feature Extraction

In machine learning, feature extraction is an important link, and the useful features extracted play a key role in the final recognition accuracy. Feature extraction can perform different types of mapping or transformation on time series signals, so as to synthesize a representative digital feature for describing time series data. In terms of emotional feature extraction, the researchers conducted research from three aspects: time domain, frequency domain and spatial information.

In the time domain, Z.Khalil *et al.* [13] extracted the standard deviation, maximum value, average value and deviation from the original EEG emotional signal, and identified positive, negative, and neutral emotions respectively, which shows that the emotion recognition of EEG more effective than other external biological information. Feradov *et al.* [14] proposed spectral EEG features based on the time-frequency analysis of multi-channel EEG signals, and the spectral power of multiple frequencies in a given range changes with time. The proposed features achieved the task of detecting emotional states with high accuracy and automatically. The emotion recognition of EEG was applied using the properties of fractal dimension by the Fraunhofer IDM group of Nanyang Technological University [15] and contrasted with other time vector EEG. The result showed that the fractal dimension showed good performance in EEG emotion recognition.

From the perspective of frequency domain characteristics, M. R. Islam *et al.* [16] proposed an EEG signal emotion recognition system based on discrete wavelet transform, which calculated wavelet energy and wavelet entropy, detected four emotions: happiness, anger, sadness and relaxation, and calculated the extracted feature values contribute to the excellent performance of the classifier. Wang Xueqin [17] used the method of multi-scale permutation entropy to extract emotional signals from EEG, thus effectively improving the recognition accuracy of emotions. Using the combination of Hilbert-Huang transform and wavelet packet transform, Yang Pengyuan [18] performed differential entropy (DE), power

spectral density (PSD) and wavelet entropy (WE) features on emotional signals in EEG for feature extraction.

From the perspective of time-frequency domain, Li Xin *et al.* [19] extracted approximate entropy, hurst exponent, wavelet coefficient energy, etc. from the four rhythmic EEG emotional signals, and used principal component analysis to analyze these features. Performed dimensionality reduction processing, and then combined the features of dimensionality reduction, and then classified EEG emotional signals. Zhuang Ning *et al.* [20] used the empirical mode decomposition method to divide EEG into five feature modes, and extracted normalized energy, first-order phase difference and first-order time difference from the five feature modes as emotion recognition features, and carried out Experimental research, results showed that first-order jet lag can better identify the emotional characteristics of brain waves.

This indicates that in recent years, academics have focused more of their attention on the three domains of time, frequency, and time-frequency. These methods all used EEG as time series as input. Machine learning algorithms may not be able to handle EEG signal sequences well because they cannot capture time series features, which have strong temporal properties and complicated features. EEG signal data is typically very voluminous and takes a lot of processing power to process. To convert EEG signals into images, image processing techniques in deep learning can be used to extract and classify EEG features. Converting EEG signals into image data can improve the efficiency and accuracy of signal processing. Using image data for classification can improve classification accuracy and reduce the workload of preprocessing.

In this experiment, two input methods were used to extract feature of EEG, and the variations between the two approaches were compared. The first was to convert the collected EEG into images, and input the images into the CNN network for feature extraction. The second was to directly preprocess the EEG and input them into the CNN network in time series for feature extraction.

C. Research status of emotion recognition based on EEG signals

Human rational decision-making, perception, learning, and other functions are significantly influenced by emotions. In order to comprehend the psychophysiological processes behind how people recognize, interpret, and imitate emotional states, emotional analysis and identification is necessary. Automatic emotion recognition has been a focus of research over the past century. Many methods of studying emotion, including facial images, gestures, language, and physiological signals, have been used. Humans cannot conceal their genuine feelings because physiological signals come from the autonomic nerve system and cannot be initiated by human conscious or unconscious control. In the last two decades, physiological signal-based emotion recognition technology has advanced quickly, and EEG-based emotion recognition research has grown exponentially. Time domain features, frequency domain features, and time-frequency domain features, which mix time domain and frequency domain, are the most often utilized EEG physiological features. These methods are therefore extremely important to the study of emotion recognition.

Many classification approaches have been presented recently in order to suit the requirements of varied data, including k-nearest neighbor (KNN) [21], decision tree (DT), random forest (RF) [22][23], and support vector machine (SVM) *et al.* [24]. EEG emotion recognition frequently used the categorization approach mentioned above.

Supported vector machine (SVM) and K closest neighbor (KNN) fusion algorithm-based emotional EEG identification model was put forth by Teng Kaidi *et al.* [25] to assess the spatial separation between the recognition sample and the ideal classification hyperplane. Used the SVM classifier if the threshold exceeded the predetermined values; otherwise, used the KNN classifier. The three categories of emotion categorization were more accurate thanks to the SVM-KNN algorithm. Subjects were instructed by Abdullah *et al.* [26] to see an image and assign it the proper emotion, with each creating a different emotional response. Used five distinct models to categorize these emotions based on valence, arousal, and dominance values. To improve classification, added new filters from Riemannian geometry. When comparing the correlation vector machine (RVM) and support vector machine (SVM), L. Xin *et al.* [27] proposed an EEG emotion recognition method based on a correlation vector machine (RVM), which will identify positive and negative emotions in the DEAP database. The classification performance of RVM was superior to that of SVM due to its lower time complexity and test error rate.

X. Deng *et al.* [28] recorded the brain waves of participants experiencing pleasant, neutral, and negative emotions using a portable EEG recorder while using 15 movies with sounds and scenes as stimulus materials. Just the 4-channel approach was used in experiments to increase classification accuracy for extra channels. In order to get more signal modes, the experiment suggested a parallel signal processing technique based on intrinsic time scale decomposition (ITD), discrete wavelet transform (DWT), variational mode decomposition (VMD), and phase space reconstruction (PSR). Eventually, the dataset yielded accuracy ratings of 90.77% and 91.38%.

M. Wu *et al.* [29] created a novel experimental methodology to examine the effectiveness of olfactory-enhanced films in evoking emotions in test subjects. Their protocol allowed odor dynamics to participate in various stages of video-evoked emotions. A total of 53 classifiers and differential entropy (DE) characteristics were used to evaluate the effectiveness of emotion recognition.

Machine learning often uses supervised learning and unsupervised learning to recognize EEG emotions. In supervised learning, training samples with well-known classification concepts are learned, and samples outside the training set are classified and predicted as much as possible. Finding structural features that can be classified from samples without a priori classification concepts is known as unsupervised learning. K-Means clustering, fuzzy clustering, and self-organizing mapping are some of the often employed techniques. Researchers have gradually begun to favor deep learning and transfer learning based on EEG classification.

To represent multi-channel EEG data using graphs and categorize EEG emotions, T. Song *et al.* [30] employed a

unique dynamic graph convolutional neural network (DGCNN) multi-channel EEG emotion detection approach. Because dynamic graph convolutional neural networks can dynamically learn the fundamental link between several EEG channels represented by matrices, they were more discriminative than conventional graph convolutional neural networks for extracting EEG features. On the SEED database, this approach had a 90.4% accuracy rate. Multi-channel EEG recordings were subjected to continuous wavelet transform by M. Islam [31] to produce wavelet coefficients in the time-frequency domain. The properties of the wavelet table graph were combined to provide the input frame for the CNN and LSTM-based neural network model. CNN-LSTM mixed network for spatiotemporal sequence feature extraction. Using modeled spatio-temporal spectral characteristics, the network performed astonishingly well in sentiment recognition for binary and multi-class classification. On the DEAP and SEED datasets, accuracy rates of 94.36% and 94.07%, respectively, were attained. A novel deep learning model called the 3D Convolutional Attention Neural Network was proposed by S. Liu *et al.* [32] (3DCANN). The network was made up of an EEG channel attention weight learning module and an out-of-control feature extraction module, which can both successfully extract the dynamic relationship between multi-channel EEG data and the internal spatial relationship of EEG signals across continuous time periods. Experiments demonstrated the 3DCANN network's superiority to the competition, which had been confirmed on the SEED database. A multi-column model built on a convolutional neural network and based on the emotion recognition module of a single convolutional neural network was proposed by Yang [33] technique enhanced by almost 10%.

Deep learning has quickly gained popularity as a research topic in recent years and has gradually replaced other methods for classifying and extracting features from EEG signals. The neural network may perform preprocessing, feature extraction, classification, and recognition when processing EEG inputs. It is possible for a neural network to finish a series of steps for processing EEG information. The flexibility of the deep neural network's structure and relatively high accuracy make it a good choice for processing EEG information.

Recurrent neural network (RNN), convolutional neural network (CNN), temporal convolutional neural network (TCN), and transfer neural network transformer are some of the specialized network topologies in neural networks that can process time series. The residual network was utilized by Rajendra Acharya *et al.* [34] to identify emotional EEG data, and it avoided the overfitting issue as well as gradient disappearance and explosion in the neural network. The accuracy rate for the three-level classification of emotions was 87.06%. Anubhav *et al.* [35] 's application of the LSTM model led to classification accuracy on the valence and arousal scales of EEG data of 94.69% and 93.13%, respectively.

In this experiment, CNN network and LSTM network were used to classify and recognize EEG signals respectively.

D. Innovations

- For the first time, the forehead stick sleep instrument was used in combination with VR headsets to monitor

the emotional state of the subjects in real time through EEG signals;

- A variety of different types of VR scenes were designed, and combined with the PANAS scale and the immersion scale to judge the correlation between the immersion and emotion of the subjects and the effect of the VR scene to stimulate emotions.

III. EMOTION INDUCTION EXPERIMENT AND DATA COLLECTION

A. Experimental design

1) Stimulus materials

a) The video selected in this experiment to inspire happiness comes from the work "Father's Funeral" in the "Annual Comedy Contest";

b) The video that stimulates the desire emotion is selected in this experiment to edit multiple eating and broadcasting videos, and the food in the video is diverse, so as to prevent a single food from being unable to stimulate the desire emotion of the testers;

c) The video selected in this experiment to stimulate healing emotions is a video of multiple cute kittens and beautiful scenery.

Three 10-second emotional marker segments are set for each video.

2) Required equipment and scales

a) EEG Recorder

The experiment used UMindSleep recorder, which is currently the smallest device for measuring EEG signals. This device measured the two electrodes FP1 and FP2 in the EEG signal, and the output result was the potential difference between the two electrodes. Therefore, this article studied the single channel of EEG.

b) VR Device

The VR device used in the experiment is Oculus QUEST2. This device uses an elastic headband, which not only adapts to the heads of different testers, but also facilitates the wearing of a frontal EEG signal recorder on the forehead.

c) PANAS Scale:

Before and after playing the video, the subjects needed to fill in this scale.

d) SAM Self-Assessment Form

In this experiment, three 10-second emotionally marked segments were selected from each emotionally stimulating video. The subjects would subjectively evaluate the video segment corresponding to the watched video, fill in the self-assessment form, and score different dimensions (1 -10 points). The scale included 7 scoring contents: valence, arousal, dominance, liking, happiness, desire and healing.

e) Immersion scale

The dependent variables used in this study were emotion, immersion, and engagement. Each option was scored on a scale of 1-5, with 1 being strongly disagree and 5 being very agree.

- Emotion: Happiness (pleasure, joy, excitement)
- Desire (desire, pursuit, satisfaction)
- Healing (relief, comfort, relief)
- I don't know what's going on around me;
- I'm aware of my surroundings;
- I feel disconnected;
- I can think of other things constantly;
- I don't feel like I'm in the real world;
- It's been a short time for me;
- I was deeply attracted by the video;

3) Participants

This experiment enrolled 36 participants, ages 20 to 30, divided into 18 female and 18 male subjects. The individuals' hearing and vision were all in normal range, and none of them had any mental or nervous system disorders. Before the experiment, informed the subjects of the experimental process and precautions. During the experiment, the participants were required to keep their bodies stable and reduce blinking or eye movement and body shaking. In order to achieve accurate measurement of EEG signals, the participants needed to wash the forehead before the experiment to avoid oil from affecting the poor contact between the equipment and the forehead.

Participants were randomly divided into two viewing conditions, one was an experimental group wearing VR to watch the video, and the other was an experimental group not wearing VR to watch the video.

4) Experimental Environment

In order to ensure that the experiment will not be disturbed by the outside world, the experiment was selected in a place with quiet sound insulation and a suitable temperature. The experimental site was equipped with a table and 2 chairs. Before the experiment, explained to the subjects how to fill in the PANAS scale and the self-assessment form report. In order to prevent the subjects from feeling dizzy due to wearing VR device, after each video was played, there was an appropriate rest and adjustment time.

5) Experimental Procedures

The experimental steps are as shown in the Figure 1.

a) First, the participants entered the experimental site, sat down and filled out the PANAS scale before watching. After completing the filling, the participants adjusted the distance between the table and the chair so that they were in a comfortable state while watching the video.

b) After wearing the forehead-stick sleep recorder and VR equipment, the participants adjusted to a comfortable state. At this time, the screen of the VR device was synchronized to the computer, so that the recorder can clearly know the screen seen by the participants.

c) Played the video that stimulated happiness emotions first. After the participants watched the video, they performed

self-assessment and scoring on the three labeled emotional segments of the video. After completing the filling, adjusted their mentality and emotions, and they can watch the next video. The procedure for testing the other two emotions is the same as this one.

d) After watching all the videos and filling out the SAM self-assessment form, filled out the PANAS and immersion scales again for after-watching evaluation.

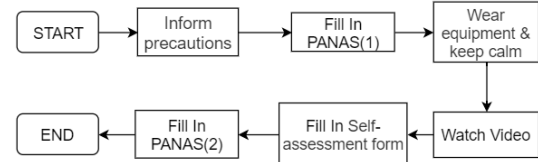


Figure 1. Experimental Procedures.

B. Survey Results and Analysis

1) Results of the PANAS scale

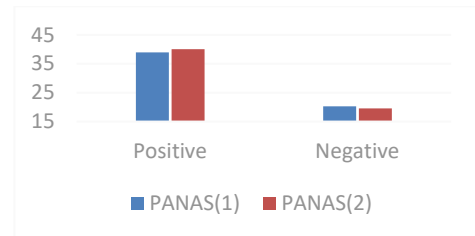
PANAS scale (1) results before watching the video:

The average value of positive emotions was 38.94, with a reliability value of 0.923, and the average number of negative emotions was 20, with a reliability value of 0.91.

PANAS scale (2) results after watching the video:

The average value of positive emotions was 40.05, with a reliability value of 0.895, and the average number of negative emotions was 19.51, with a reliability value of 0.901.

TABLE I. PANAS (1) (2) DIFFERENCES



As shown in the TABLE I, after watching the play video, the positive emotions of the test subjects increased, and the negative emotions also decreased.

2) Results of the SAM scale

The average values and reliability values of each dimension of the three emotions of the SAM scale were as follows:

TABLE II. THE AVERAGE VALUE AND RELIABILITY VALUE OF EACH DIMENSION IN THE SAM TABLE

	Happy	Desire	Healing
Valence	7.27	5.48	5.59
Arousal	6.51	6.15	5.36
Dominance	6.4	5.37	6.4
Liking	7.42	6.26	7.36
Alpha	0.86	0.853	0.787

TABLE III. CORRELATION COEFFICIENTS CORRESPONDING TO PAIRS OF 7 DIMENSIONS

	Valence	Arousal	Dominance	Liking	Happy	Desire	Curative
Valence	1						
Arousal	0.85**	1					
Dominance	1.36*	-0.643	1				
Liking	0.843**	0.565*	-0.294	1			
Happy	0.864**	0.23**	0.352**	0.776**	1		
Desire	0.376*	0.786*	0.235**	0.432*	0.342**	1	
Curative	-0.127	0.165**	0.25	0.33	-0.75*	-0.26	1

It can be seen from the above TABLE II that the reliability value of happiness was 0.86, the reliability value of desire was 0.853, and the reliability value of healing was 0.787, all of which exceeded 0.7, so it showed that the correlation between the data was high, and this table also had certain effectiveness.

The corresponding correlation coefficients among the seven dimensions were shown in the TABLE III.

It can be seen from the TABLE III that there was a significant correlation between valence and happiness ($p < 0.001$) and desire ($p < 0.05$). Arousal had a significant correlation with happiness ($p < 0.001$), desire ($p < 0.05$), healing ($p < 0.001$). There was a significant correlation between dominance and happiness ($p < 0.05$), desire ($p < 0.001$). There was a significant correlation between liking and happiness ($p < 0.001$), desire ($p < 0.050$). happiness-desire and happiness-healing had a significant correlation.

The correlation between the three emotions was not so significant, but it had been proved by experiments that the three emotions can also be well identified and classified.

3) Result of Immersion scale

The results of the immersion scale are shown in TABLE IV.

a) Happiness

The VR-wearing group had higher emotional responses than the non-VR-wearing group.

The immersion feeling of the wearing VR group was higher than that of the non-VR group.

The sense of immersion was significantly correlated with the happiness of the participants.

b) Desire

The VR-wearing group had higher emotional responses than the non-VR-wearing group.

The immersion feeling of the wearing VR group was higher than that of the non-VR group.

The correlation between immersion and desire emotions of the participants was not significant.

c) Healing

The VR-wearing group had higher emotional responses than the non-VR-wearing group.

The immersion feeling of the wearing VR group was higher than that of the non-VR group.

The sense of immersion was significantly correlated with the healing emotions of the participants.

TABLE IV. RESULT OF IMMERSION SCALE

Emotion	Item	Group	M	SD	
Happiness	Relative Emotion	VR	3.87	0.88	T=2.13
		Non-VR	3.13	0.79	P=0.027
	Immersion	VR	3.43	0.79	T=2.08
		Non-VR	3.02	0.77	P=0.032
Desire	Relative Emotion	VR	2.89	1.12	T=0.79
		Non-VR	2.77	1.44	P=1.225
	Immersion	VR	2.65	1.44	T=1.19
		Non-VR	2.45	0.77	P=1.03
Healing	Relative Emotion	VR	3.66	0.964	T=1.96
		Non-VR	3.48	1.12	P=0.04
	Immersion	VR	3.75	0.54	T=1.52
		Non-VR	3.24	1.64	P=0.035

IV. EMOTION RECOGNITION WITH CNN NETWORK

A. Dataset and Experimental Environment

The data set was as mentioned in the previous section. There were a total of nine video marker segments, a total of 36 people were tested, and a total of 315 EEG images, as shown in the Figure 2. The experimental environment was python3.9, TensorFlow, pytorch==1.13.1. The processor was AMD Ryzen 7 5800H with Radeon Graphics.

B. Preprocessing

The collected EEG signals were taken as input in the form of pictures. As shown in the Figure 2, the picture was the original form of the EEG signal. After preprocessing, there were 300 pictures. The ability of neural networks to automatically learn complex features from a large amount of data advocates that researchers can directly input raw EEG signals into neural networks. After the data set was opened by EDFbrowser, selected the channel of EEG, because the channels measured by this device also include pulse, blood oxygen, body posture, etc. Adjusted the interface to the position of the EEG signal corresponding to the selected emotional marker segment, and adjusted the timescale and amplitude. At this time, the EEG can be displayed more clearly. Finally printed the image format. After the network read the picture file, it performed transforms.ToTensor() processing on it. This step was to convert the picture into Tensor format. This format can be directly input into the neural network, and then transforms.Normalize() was performed on the pixel value.

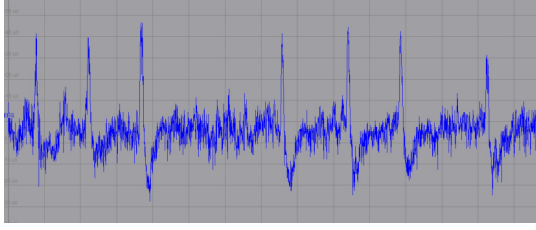


Figure 2. Raw EEG image.

C. Feature Extraction

The Resnet18 network was utilized in this experiment. The Resnet18 network is based on the CNN network, with an additional step of a residual block. The Resnet network propagates forward while propagating the error backward, unlike a regular neural network.

The forward propagation is a direct forward process, and the operation of continuous addition is adopted (the operation of multiplying the convolutional layer in the residual block). It can solve the problem of gradient disappearance. In the forward convolution, each layer of convolution extracts only a part of the features of the previous layer image. The deeper the network layer, the more original image information is lost, which is prone to underfitting. At this time, it is necessary to add a shortcut structure, which is equivalent to adding all the information of the previous layer image to each block, and to a certain extent reduce the loss to a lower level.

The Resnet18 network has five convolution groups, the first convolution group includes only one convolution calculation, and the second to fifth groups have multiple residual blocks. The Resnet18 network structure is shown in the Figure 3.

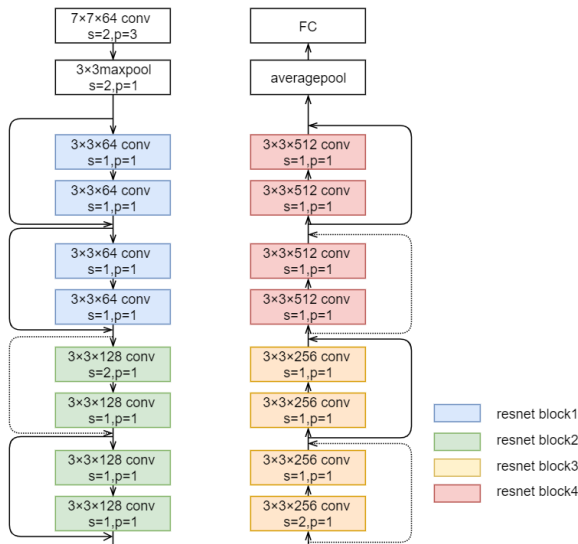


Figure 3. Resnet18 network architecture.

The residual block's responsibility is to accept the output of the preceding layer as an input. Compare the input and output outcomes following the convolution computation in the residual block. If things improve, the residual block is required. If not, the residual is disregarded piece. It can be summed up in

one word: "at least not bad." The Figure 4 depicts the remaining block's structural layout.

After the preprocessed image is extracted by the neural network, the feature vector is obtained.

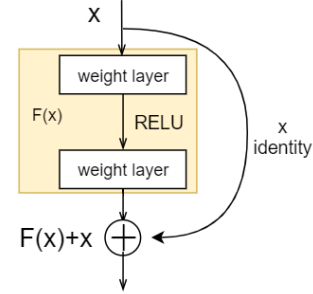


Figure 4. Residual Network Architecture.

D. Emotion recognition classification

The loss function used in the experiments is the multi-classification cross-entropy loss function.

$$\text{Loss} = -\sum_{i=1}^K y_i \log(p_i) \quad (1)$$

K was the number of categories, y was the label, and p was the output of the neural network.

The optimizer used in the experiment was the torch.optim.Adam() method. Experiments adjusted the lr learning rate to 1e-5. Because lr was the learning rate, also known as the step size factor, if the learning rate was too low, the learning speed would be too slow, and if the learning rate was too high, it would be difficult to converge. Lr controlled the update ratio of the weights. The larger the value, the faster the initial learning before the learning rate was updated. The smaller the value, the better the training convergence and found the optimal solution. The experiment set the learning rate to 1e-5, which can achieve satisfactory results. The value of weight decay was set to 1e-5. Weight decay was essentially an L2 regularization coefficient, adding this L2 regularization would limit the weight of the model to tend to 0. Weight decay can prevent the problem of overfitting, kept the weight at a small value, and avoided gradient explosion.

E. Training and Test Results

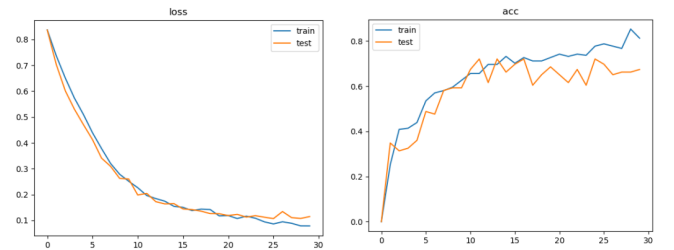


Figure 5. Accuracy and Loss.

As shown in the Figure 5, the experimental data set was divided into training set and test set according to the ratio of 7:3. The final accuracy on the test set was 0.93.

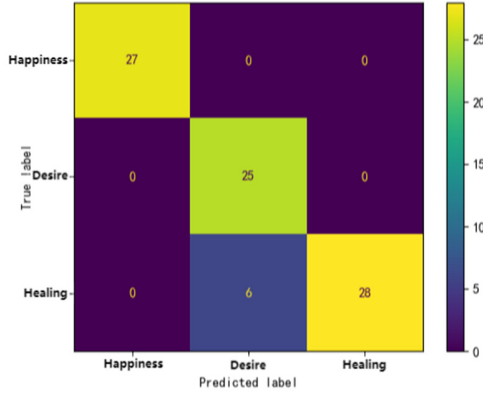


Figure 6. Forecast result.

The visualization of the final recognition result was shown in the Figure 6. It can be seen that the happy emotion can be fully recognized.

V. EMOTION RECOGNITION WITH CNN AND LSTM

A. Data Preprocessing

Training set and test set were split up into the data set in an 8:2 ratio. To obtain a one-dimensional vector from the data set in edf format, used Python's pyedflib library. Then encoded the vector with one_hot classification to vectorize the label. Then normalized the numbers.

B. Feature extraction

Constructed a one-dimensional CNN neural network, which included 1 word embedding layer, 2 convolutional layers, 1 maximum pooling layer and 1 fully connected layer. First, the data reduced the high-dimensional one-hot through the word embedding layer, because when we one-hot processed the data, the vector would become high-dimensional and sparse. Then entered the Dropout layer to prevent the model from overfitting and improve the generalization ability of the model. The convolutional layer extracted the features through convolution, and then reduced the dimensionality of the features through the pooling layer. When only one layer of convolution was used, the effect of feature extraction was not ideal, but the effect was significantly improved after adding a layer. The value output by the fully connected layer was judged by the sigmoid function which type of emotion it was.

C. Emotion recognition classification

The LSTM long short-term memory network was used in the experiment to recognize and categorize emotions.

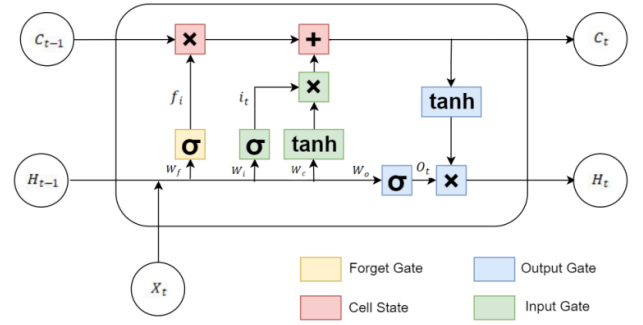


Figure 7. LSTM architecture

The forgetting gate, represented by the yellow block, erased insignificant sequences and determines whether they were worthwhile. The sigma function took the input data and the data from the previous concealed state as inputs; it output a value between 0 and 1, discarded values close to 0 and kept values close to 1.

The green block served as the input gate, which computed the input and hidden state, and updated the state with the obtained result.

The blue block was the output gate, which transferred the output of the previous frame and the content of its own input to the sigma function, and transferred the obtained new cell state to the tanh function. These two values were multiplied to obtain the information of the hidden state, and the information was used as the output of the current cell, and the new cell state was output.

As shown in the Figure 7, x_t referred to the feature sequence input at time t , C_{t-1} represented the cell state output by the previous unit, and H_{t-1} represented the hidden state output by the previous unit. f_t was the information output by the forget gate, and W_f was the connection weight of the forget gate. The information output by the input gate, its expression is:

$$f_t = \sigma(W_f \cdot [H_{t-1}, x_t] + b_f) \quad (2)$$

Among them, b_f is the bias vector of the forget gate.

$$i_t = \sigma(W_i \cdot [H_{t-1}, x_t] + b_i) \quad (3)$$

$$\tilde{C}_t = \tanh(W_c \cdot [H_{t-1}, x_t] + b_f) \quad (4)$$

Equation (3) (4) was the expression of the input gate. where W_i and W_c were the connection weights associated with the input gates.

$$o_t = \sigma(W_o \cdot [H_{t-1}, x_t] + b_o) \quad (5)$$

$$H_t = o_t * \tanh(C_t) \quad (6)$$

The above formula was the output gate expression. Where W_o was the connection weight associated with the output gate and b_o was the bias vector of the output gate.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (7)$$

The above formula represented the update of the cell state.

D. Training and Test Results

As shown in the Figure 8, the accuracy rate of this experiment can reach 94.9%.

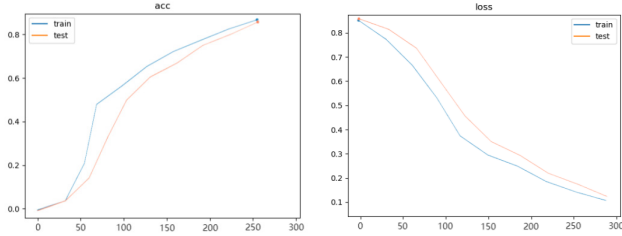


Figure 8. Accuracy and Loss.

In the classification of predicted emotions, happiness emotions can also be better identified as shown in Figure 9.



Figure 9. Forecast result.

VI. SUMMARY AND OUTLOOK

Based on the relevant steps of EEG emotion recognition, this paper summarized some common emotion recognition technologies at home and abroad, and compared the advantages and disadvantages of each method. This paper used EEG signal-based emotion recognition and compared the effects of CNN and LSTM neural network recognition. This article used the PANAS scale and the self-assessment report to analyze the emotional state of the subjects. And through the design of emotion-induced experiments, the EEG signals of 35 people under the conditions of happiness, desire and healing were collected, and the EEG signals were preprocessed, feature extracted and classified and recognized. Compared the difference between EEG signals input into the CNN network in the form of pictures and in the form of vectors for feature extraction, and compared the accuracy of emotion classification and recognition using CNN and LSTM networks. The accuracy rate of classification with CNN network reached 93.6%, and the accuracy rate of classification with LSTM network reached 94.9%.

Because of the power of the neural network, only one neural network can be used to operate all the steps of processing EEG signals, and finally can achieve very good results. This also allows subsequent researchers to integrate multiple neural networks to study EEG signals. At present, many experiments are to distinguish positive, neutral and negative emotions, or to induce negative emotions and study them. There are still very few studies on positive emotion recognition. As far as life is getting more and more stressful, we should do more research on positive emotions, and we can invent some software that induces and recognizes positive emotions. As far as the emotion-related EEG database is concerned, the number of databases is small, the number and types of test subjects are not many, and the emotions they have are not many. Future work on collecting EEG datasets about emotions could be implemented.

ACKNOWLEDGE

The funding project of this research is: Macau Foundation (Macau Foundation) scientific research project "Research on Obstacles, Strategies and Technologies of People's Livelihood Data Cross-Border in the Greater Bay Area: Taking Medical and Health Care Data Cross-Border as an Example" (Grant No.MF2102), Macau

REFERENCES

- [1] Savran A, Ciftci K, Chancel G, et al. Emotion Detection in the Loop from Brain Signals and Facial Images[J]. Acta Horticulturae, 2016, 671(671):151-157.
- [2] J. A. Healey and R. W. Picard, "Detecting stress during real-world driving tasks using physiological sensors," in IEEE Transactions on Intelligent Transportation Systems, vol. 6, no. 2, pp. 156-166, June 2005, doi: 10.1109/TITS.2005.848368.
- [3] S. Koelstra et al., "DEAP: A Database for Emotion Analysis ;Using Physiological Signals," in IEEE Transactions on Affective Computing, vol. 3, no. 1, pp. 18-31, Jan.-March 2012, doi: 10.1109/T-AFCC.2011.15.
- [4] M. Pantic, M. Valstar, R. Rademaker and L. Maat, "Web-based database for facial expression analysis," 2005 IEEE International Conference on Multimedia and Expo, Amsterdam, Netherlands, 2005, pp. 5 pp.-, doi: 10.1109/ICME.2005.1521424.
- [5] Valstar M, Pantic M. Induced disgust, happiness and surprise: An addition to the mmi facial expression database[J]. Proc.intern.workshop on Emotion Corpora for Research on Emotion & Affect, 2010:65-70.
- [6] Douglas-Cowie E, Cowie R, Schrder M. A New Emotion Database: Considerations, Sources and Scope[C]. ISCA Workshop on Speech and Emotion, 2020:39-44.
- [7] Siedlecka, E., & Denson, T. F. (2019). Experimental Methods for Inducing Basic Emotions: A Qualitative Review. Emotion Review, 11(1), 87-97. <https://doi.org/10.1177/1754073917749016>
- [8] Z. Li, M. Li, C. He, H. Chen, Y. Wang and Z. Li, "Pattern Analysis of Stimuli-Induced and Self-Induced Emotions Based on EEG," 2022 5th International Conference on Pattern Recognition and Artificial Intelligence (PRAI), Chengdu, China, 2022, pp. 870-874, doi: 10.1109/PRAI55851.2022.9904010.
- [9] S. Jaidee, K. Wongpatikaseree, N. Hnoohom, S. Yuenyong and P. Yomaboot, "MU Face Emotion - Building a Large Dataset for Emotional Facial Expression in Psychological Domain," 2022 14th International Conference on Knowledge and Systems Engineering (KSE), Nha Trang, Vietnam, 2022, pp. 1-6, doi: 10.1109/KSE56063.2022.9953783.
- [10] S. Koelstra et al., "DEAP: A Database for Emotion Analysis ;Using Physiological Signals," in IEEE Transactions on Affective Computing, vol. 3, no. 1, pp. 18-31, Jan.-March 2012, doi: 10.1109/T-AFCC.2011.15.
- [11] A. Gruenewald et al., "[Regular Paper] Biomedical Data Acquisition and Processing to Recognize Emotions for Affective Learning," 2018 IEEE 18th International Conference on Bioinformatics and Bioengineering

- (BIBE), Taichung, Taiwan, 2018, pp. 126-132, doi: 10.1109/BIBE.2018.00031.
- [12] Bos D O. EEG-based emotion recognition[J]. The influence of visual and auditory stimuli, 2006, 56(3): 1-17.
- [13] Z. Khalili and M. H. Moradi, "Emotion recognition system using brain and peripheral signals: Using correlation dimension to improve the results of EEG," 2009 International Joint Conference on Neural Networks, Atlanta, GA, USA, 2009, pp. 1571-1575, doi: 10.1109/IJCNN.2009.5178854.
- [14] Feradov F N, Ganchev T D. Spectral Features of EEG Signals for the Automated Recognition of Negative Emotional States[C]//2019 IEEE XXVIII International Scientific Conference Electronics (ET). IEEE, 2019: 1-4.
- [15] Ansari-Asl K, Chaneel G, Pun T. A channel selection method for EEG classification in emotion assessment based on synchronization likelihood[C]//2007 15th European signal processing conference. IEEE, 2007: 1241-1245.
- [16] M. R. Islam and M. Ahmad, "Wavelet Analysis Based Classification of Emotion from EEG Signal," 2019 International Conference on Electrical, Computer and Communication Engineering (ECCE), Cox'sBazar, Bangladesh, 2019, pp. 1-6, doi: 10.1109/ECACE.2019.8679156.
- [17] Wang Xueqin. Analysis and Research on Emotion Recognition Based on Multiple EEG Signal Features [D]. Nanjing University of Posts and Telecommunications, 2020. DOI: 10.27251/d.cnki.gnjdc.2020.000098.
- [18] Yang Pengyuan, Li Haifang, Chen Dongwei. Application of Hilbert-Huang Transformation in EEG Feature Extraction [J]. Computer Engineering and Design, 2014,35(07):2509-2514.DOI:10.16208/j.issn1000- 7024.2014.07.019.
- [19] Li Xin, Cai Erjuan, Tian Yanxiu, Sun Xiaoqi, Fan Mengdi. An improved EEG feature extraction algorithm and its application in emotion recognition [J]. Journal of Biomedical Engineering, 2017,34(04):510-517+ 528.
- [20] Zhuang N, Ying Z, Tong L, et al. Emotion Recognition from EEG Signals Using Multidimensional Information in EMD Domain[J]. BioMed research international, 2017, 2017:8317357.
- [21] Jadhav N, Manthalkar R, Joshi Y. Electroencephalography-based emotion recognition using gray-level co-occurrence matrix features[C]//Proceedings of International Conference on Computer Vision and Image Processing: CVIP 2016, Volume 1. Springer Singapore, 2017: 335-343.
- [22] Li C, Xu C, Feng Z. Analysis of physiological for emotion recognition with the IRS model[J]. Neurocomputing, 2016, 178: 103-111.
- [23] Zheng X, Zhang M, Li T, et al. A novel consciousness emotion recognition method using ERP components and MMSE[J]. Journal of Neural Engineering, 2021, 18(4): 046001.
- [24] Zhang Y, Ji X, Liu B, et al. Combined feature extraction method for classification of EEG signals[J]. Neural Computing and Applications, 2017, 28(11): 3153-3161.
- [25] Teng Kaidi, Zhao Qian, Tan Haoran, Zheng Jinhe, Dong Yixian, Shan Hongfang. Emotional EEG Recognition Based on SVM-KNN Algorithm[J]. Computer System Application, 2022,31(02):298-304.DOI:10.15888/j.cnki.csa.008332.
- [26] M. A. Abdullah and L. R. Christensen, "EEG Emotion Detection Using Multi-Model Classification," 2018 International Conference on Bioinformatics and Systems Biology (BSB), Allahabad, India, 2018, pp. 178-182, doi: 10.1109/BSB.2018.8770695.
- [27] L. Xin, S. Xiao-Qi, Q. Xiao-Ying and S. Xiao-Feng, "Relevance Vector Machine Based EEG Emotion Recognition," 2016 Sixth International Conference on Instrumentation & Measurement, Computer, Communication and Control (IMCCC), Harbin, China, 2016, pp. 293-297, doi: 10.1109/IMCCC.2016.106.
- [28] X. Deng, X. Lv, P. Yang, K. Liu and K. Sun, "Emotion Recognition Method Based on EEG in Few Channels," 2022 IEEE 11th Data Driven Control and Learning Systems Conference (DDCLS), Chengdu, China, 2022, pp. 1291-1296, doi: 10.1109/DDCLS55054.2022.9858390.
- [29] M. Wu, W. Teng, C. Fan, S. Pei, P. Li and Z. Lv, "An Investigation of Olfactory-Enhanced Video on EEG-Based Emotion Recognition," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 31, pp. 1602-1613, 2023, doi: 10.1109/TNSRE.2023.3253866.
- [30] T. Song, W. Zheng, P. Song and Z. Cui, "EEG Emotion Recognition Using Dynamical Graph Convolutional Neural Networks," in IEEE Transactions on Affective Computing, vol. 11, no. 3, pp. 532-541, 1 July-Sept. 2020, doi: 10.1109/TAFFC.2018.2817622.
- [31] M. Islam and T. Lee, "Wavelet based Emotion Detection from Multi-channel EEG using a Hybrid CNN-LSTM Model," TENCON 2022 - 2022 IEEE Region 10 Conference (TENCON), Hong Kong, Hong Kong, 2022, pp. 1-6, doi: 10.1109/TENCON55691.2022.9978122.
- [32] S. Liu et al., "3DCANN: A Spatio-Temporal Convolution Attention Neural Network for EEG Emotion Recognition," in IEEE Journal of Biomedical and Health Informatics, vol. 26, no. 11, pp. 5321-5331, Nov. 2022, doi: 10.1109/JBHI.2021.3083525.
- [33] Yang H, Han J, Min K. A Multi-Column CNN Model for Emotion Recognition from EEG Signals. Sensors (Basel). 2019 Oct 31;19(21):4736. doi: 10.3390/s19214736. PMID: 31683608; PMCID: PMC6865186.
- [34] U. Rajendra Acharya, Shu Lih Oh, Yuki Hagiwara, Jen Hong Tan, Hojjat Adeli, Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals, Computers in Biology and Medicine, Volume 100, 2018, Pages 270-278, ISSN 0010-4825, https://doi.org/10.1016/j.compbiomed.2017.09.017.
- [35] Anubhav, D. Nath, M. Singh, D. Sethia, D. Kalra and S. Indu, "An Efficient Approach to EEG-Based Emotion Recognition using LSTM Network," 2020 16th IEEE International Colloquium on Signal Processing & Its Applications (CSPA), Langkawi, Malaysia, 2020, pp. 88-92, doi: 10.1109/CSPA48992.2020.9068691.